

Onatola Timilehin Faruq

Smart Contract Security Engineer | Oracle Testing | Automated Audit Tooling

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Summary

Security-focused blockchain engineer with production experience testing oracle-dependent DeFi protocols. Built Sentinel, a multi-agent automated smart contract auditing framework with Slither/Mythril integration and Foundry PoC generation. Established testing infrastructure achieving 95%+ coverage across Aave v3, Uniswap, Curve, and GMX integrations. Hands-on with Chainlink oracles, ZK proofs, and TEE-based privacy systems.

Technical Expertise

Testing & Security: Foundry (Forge, fuzz testing, fork testing) | Hardhat | Slither | Mythril | Semgrep | Security Auditing | Stress Testing | Fault Tolerance Testing

Smart Contracts: Solidity (Advanced) | Rust/Anchor (Intermediate) | Noir | Clarity

Oracle Systems: Chainlink Data Feeds | Chainlink VRF | Chainlink Automation | Chainlink CCIP | Multi-keeper consensus | Data staleness detection

Scripting & Automation: TypeScript | Python | JavaScript | Ethers.js | Viem | Web3.js

Privacy/ZK: Zero-Knowledge Proofs (Noir/UltraHonk) | TEE (Trusted Execution Environment) | Stealth Addresses

Infrastructure: GitHub Actions CI/CD | Docker | Tenderly | The Graph | PostgreSQL | Redis

Chains: Ethereum | Base | Arbitrum | Optimism | BNB Chain | Solana | Hedera

Professional Experience

Blockchain Engineer | DeFiConnectCredit | Jan 2025 - Present

- Established Foundry testing framework with **95%+ code coverage** across multi-protocol integration suite (Aave v3, Uniswap v3/v4, Curve, GMX)
- Identified and resolved **3 critical vulnerabilities** pre-deployment through systematic security review, including oracle manipulation and reentrancy vectors
- Built stress testing suite simulating high-gas conditions, network latency, and protocol edge cases across 4 DeFi protocols
- Reduced transaction costs by **35%** through gas optimization, validating improvements via comparative benchmarking
- Created deployment automation suite with built-in verification checks, cutting deployment time from 2 hours to 15 minutes

Blockchain Engineer (Intern) | DeFiConnectCredit | Jun 2024 - Dec 2024

- Conducted security reviews on **15+ contracts** pre-deployment, identifying reentrancy, front-running, and oracle manipulation vulnerabilities
- Developed test suites with **90%+ coverage** including edge case testing, invariant testing, and stress testing
- Built and tested ERC721 incentive system with Merkle tree proofs for 500+ users
- Contributed to collateral management and liquidation mechanism testing for core lending protocol

Security Tooling

Sentinel - Automated Smart Contract Auditing Framework

Multi-agent security auditor designed for competitive audit contests (Sherlock, Code4rena, Cantina)

- Architected multi-agent pipeline: reconnaissance, static analysis (Slither/Mythril), LLM-powered deep analysis, PoC generation
- Built specialized vulnerability hunters for **oracle manipulation**, flash loan attacks, reentrancy (classic, cross-function, cross-contract, read-only), and access control
- Implemented automated Foundry-based PoC generation with mainnet fork testing
- Supports 4 languages: Solidity, Rust/Solana, Move (Aptos/Sui), Cairo/StarkNet
- Integrated Semgrep rules for static pattern matching with cost-aware budget allocation

Python Slither Mythril Foundry Semgrep

Relevant Projects

Sentiment - Uniswap v4 Dynamic Fee Hook

UHI7 Incubator Capstone | Oracle-dependent dynamic fee hook consuming data from 8 off-chain sources

- Tested multi-keeper architecture (Chainlink Automation + Gelato) for redundancy and failover
- Validated data feed accuracy, staleness detection, and consensus across multiple oracle sources
- Achieved **<30k gas overhead** per swap through optimization and gas benchmarking
- Tested fee adjustment algorithm (0.25%-0.44%) under simulated market stress conditions

Solidity Foundry Chainlink Uniswap v4

Nocturne - Private Payments on Solana

ZK Privacy Protocol | Directly relevant to privacy-related oracle technologies (ZK + TEE)

- Implemented and tested Noir/UltraHonk zero-knowledge proof circuits for withdrawal verification
- Built TEE relay for identity protection with fault tolerance testing
- Validated Merkle commitment pools and stealth address (DKSAP) implementation

- Achieved **97% unlinkability score** through systematic privacy testing

Rust Anchor Noir Zero-Knowledge TEE

TruthBounty - Prediction Market Reputation Protocol

2nd Place, Seedify Hackathon | Oracle-integrated reputation system across 13 markets

- Built data verification pipeline across Polymarket, Azuro, Thales, and 10 other prediction platforms
- Tested soulbound NFT minting with on-chain SVG generation and custom subgraph indexing
- Validated Wilson Score confidence interval algorithm against edge cases

Solidity The Graph Soulbound NFTs Next.js

Certifications & Training

- **Uniswap Hook Incubator (UHI7)** - Graduate (15% acceptance rate)
- **Cyfrin Updraft** - DeFi Security & Protocol Engineering (Aave v3, Uniswap, GMX, Curve, Auditing)
- **School of Solana** - Ackee Blockchain (Rust/Anchor Development)
- **Hashgraph Developer Certification** - The Hashgraph Association

Education

BSc Physiotherapy | University of Ibadan, College of Medicine | 2024

Portfolio: faruukku.vercel.app | GitHub: github.com/Leihyn | References available upon request